Pulse + Petals: Biomimetic Fas

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Abstract

This interdisciplinary project explores fashion technology
the creation of a bio-mimetic and bio-responsive dress. Thi
mechanisms which are controlled by real-time PPG heart rate
physiological state and translates it into expression thro
tween the two. The technical methodology integrates robotic
a MAX30102 pulse oximeter connected to a ESP32-C3 microcont
heart rate data taken from the MAX30102 and transmits it wire
ceiver" ESP32-C3 microcontroller. The receiver is hidden an
Two 3D printed fower mechanisms which are controlled by one se
are embedded into the dress. Teensy 4.1 then coordinates the
the fower mechanisms. When the user's heart rate is consider
bloom in a slow, calmer motion, while higher heart rates (100
motions. This range of excitement creates a representation o
nonverbal communication through bio-responsive fashion. Ov
ical design and interconnected sensor networks that can tra
project contributes to emerging felds of afective computing
ogy's potential to become interconnected with fashion and hu

1 Introduction

Wearable technology has evolved signifcantly over the past tw
ers and medical monitoring devices into increasingly comple
environmental interaction, and computational adaptation.
wearable technology continues to prioritize technical func
in devices that are utilitarian rather than emotionally exp
electronic textiles demonstrated the potential for garments
taining the softness, fexibility, and [1] However, the
of embedded electronics into fashion remains constrained by
cohesion between technological systems and garment design [2]

Within computational fashion, designers such as Anouk Wippr
formative potential of bio-responsive Spider Dress through pro
couture [3]. These garments integrate sensors, actuators, an
that physically responds to human presence, environmental
boundary between fashion design and machine behavior. More
brain-computer-interface-driven garments have further exp
ical and cognitive signals can be translated [4][5] into dynamic ar

At the same time, advances in biometric sensing have made it i
indicators such as heart rate, pulse waveforms, and cardiov

Photoplethysmography (PPG)-based sensors—which sense hear
infrared light—have shown strong potential for continuous b
health-state detection in wearable applications [6][7][8]. Among physiological signals, heart r
particularly compelling input for emotionally responsive de
are associated with changes in autonomic nervous system acti
[9][10]. While these sensing systems have been extensively exp
monitoring applications, their use as a mechanism for artist
limited.

A signifcant barrier in bio-responsive fashion is the physi
into wearable structures. Materials such as textiles, ther
structural components impose engineering constraints rela
and mechanical stability. Recent advances in 3D-printed tex
fabrication have begun addressing these limitations [11], enabl
[12][13]. However, the challenge of combining biometric sensing
and couture garment construction into a cohesive, aesthetic
unexplored.

Therefore, this study investigates the following research q
ded robotic systems be integrated into wearable fashion to c
physiological state through dynamic, aesthetically meaning

To address this question, this research developed a bio-res
cardiovascular activity into kinetic foral motion. The impl
plethysmography (PPG) heart rate sensor, dual ESP32-C3 mic
NOW wireless networking, and a Teensy 4.1 microcontroller r
fower mechanisms. Mechanically, the system is integrated in
tural brackets designed to support and elevate the fower asse
the dress. By combining biometric sensing, embedded communi
turing, and couture garment construction, this project expl
expressive interface between human physiology, machine resp

2 Procedure

2.1 Design Software and Fabrication Environment

The mechanical components of the wearable system were design
(CAD) modeling was used to develop both the foral actuator ho
required for garment integration. Iterative prototyping al
ment points to be refned prior to physical production.

All mechanical components, including the fower mechanisms a
tured through additive manufacturing using 3D printing. Thi
control while allowing lightweight structural components t
tion.

Figure 1: Initial petal, ring, and rack prototypes fo

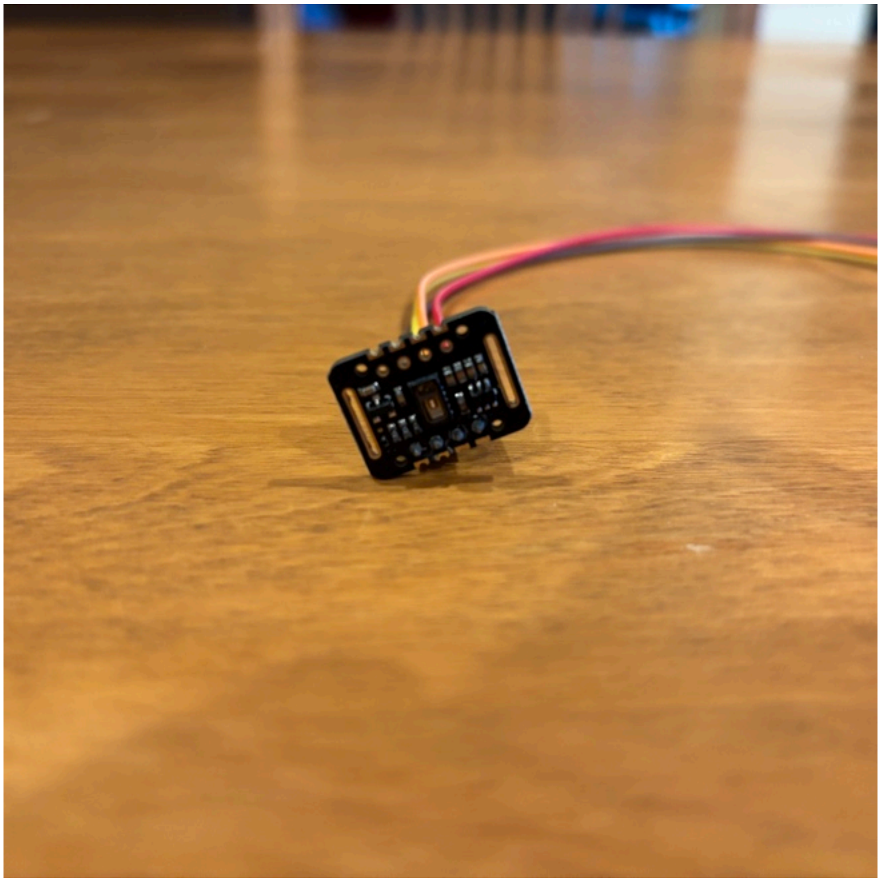
Embedded system programming was conducted using Arduino IDE

The table below summarizes the primary hardware components u

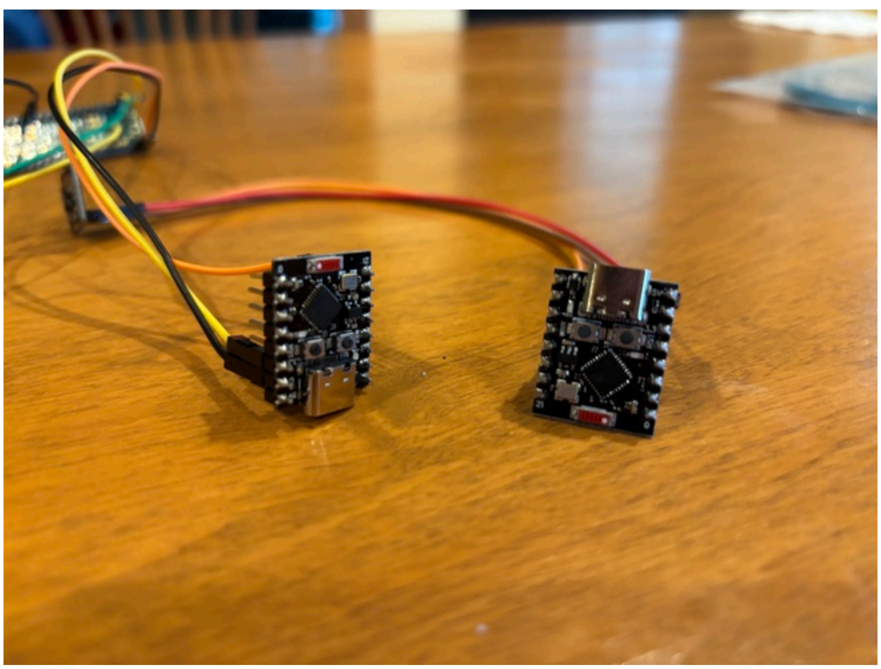
Component	Purpose
MAX30102	Heart rate and pulse sensing
ESP32-C3 (2x)	Wireless transmission and reception
Teensy 4.1	Motion control and actuation
Servo motors	Flower movement
Rechargeable battery	Portable power supply

Primary hardware components used in prototype development a

Figure 2: Miuzei servo used for prototype devel



F i g u r e 3 : MAX30102 PPG h e a r t s e n s o r u s e d f o r c o l l e c t



F i g u r e 4 : T w o ESP32 C3 M i c r o c o n t r o l l e r s u s e d f o r w i r e

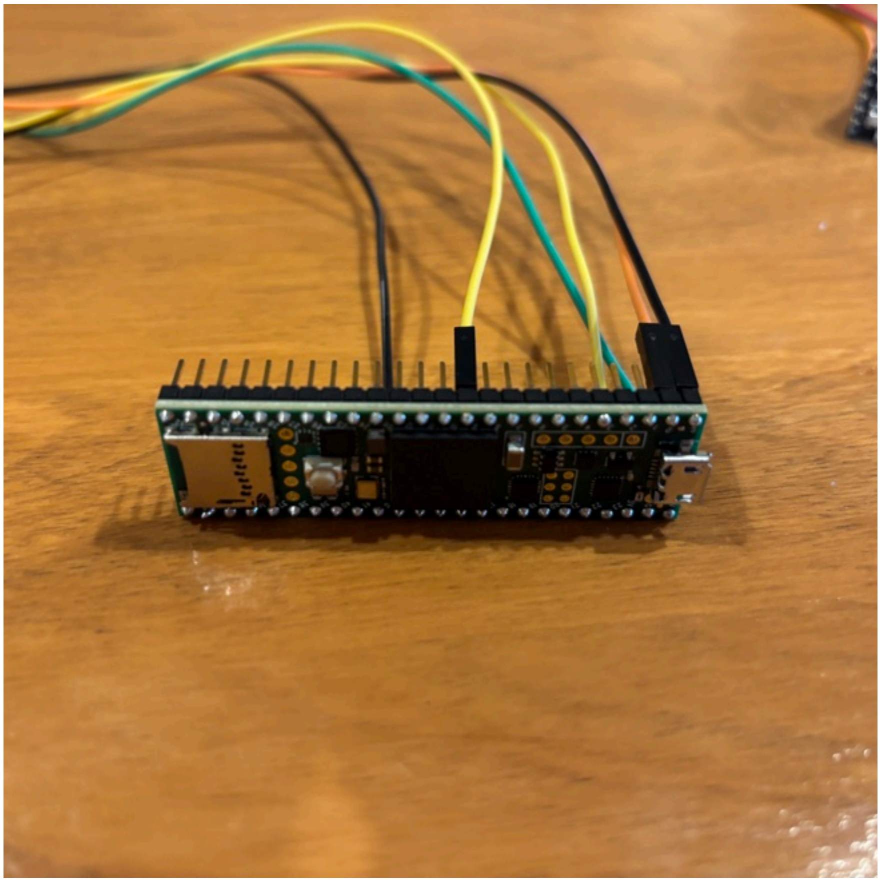


Figure 5: Teensy 4.1 used as main microcontrol

2.2 Sensor Validation and Embedded Communication

System development began with validation of the MAX30102 ph
rate acquisition testing was performed using an Arduino Uno
full system integration. Sensor output was monitored throug
consistency and signal stability.

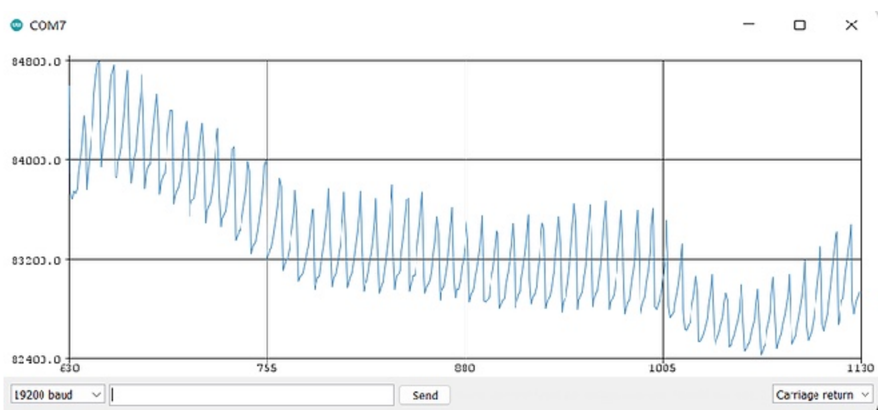


Figure 6: Output of MAX30102 heart monitor when connec

Following successful biometric sensing, wireless communica
crocontrollers using the ESP-NOW communication protocol. O
mitter responsible for collecting sensor data, while the sec
trials were conducted to verify packet delivery and communic

After successful wireless validation, the receiver ESP32-C3
The Teensy interpreted incoming biometric data and generate
rate fuctuations.

2.3 Mechanical Design and Garment Construction

The foral actuation mechanism was designed in Fusion 360 and
actuation mechanism chosen was the rack and pinion. Each pe
to a ring and be meshed by the center of the fower mechanism wh
horn was modeled to ensure smooth meshing, and was attached
ensure aesthetics. The mechanism and casing printed in PLA p
The fower geometry was optimized to balance aesthetic movem
constraints. The mass of the 3D printed component weighed 51

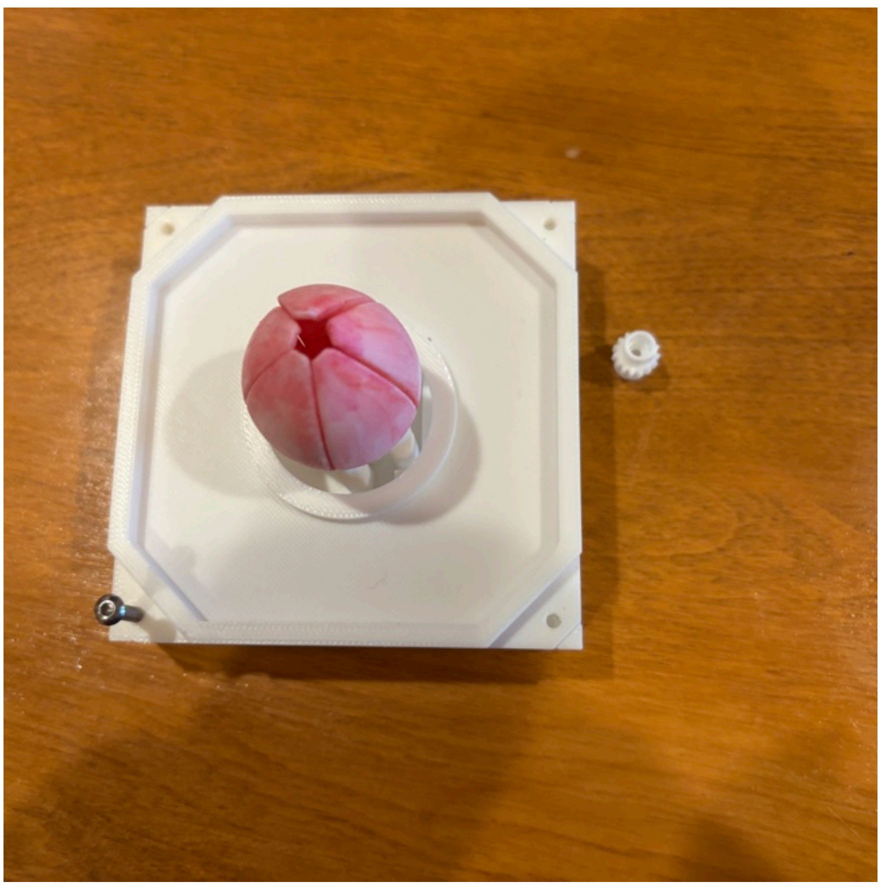


Figure 7: Final Prototype of Flower Mechanism

A secondary structural bracket was also modeled in Fusion an
of the fower mechanism during wear. This bracket served as t
assembly and the garment structure.

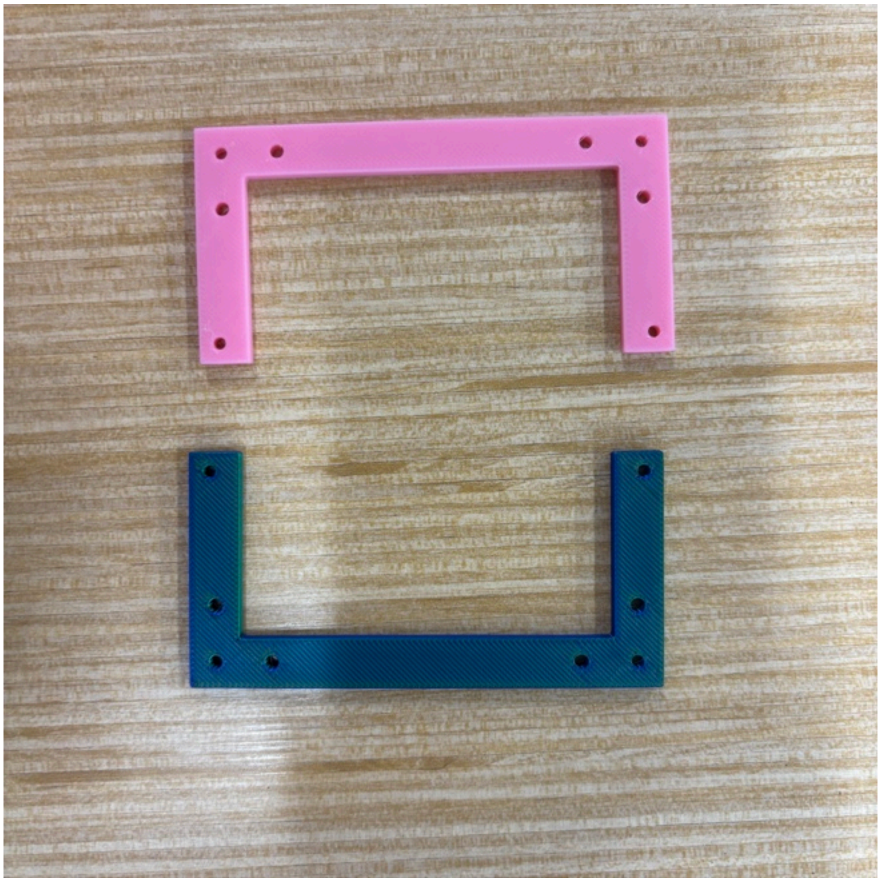


Figure 8: 3D printed brackets used for attaching fowe

The garment foundation was constructed using a corset-style bracket assembly was sewn directly into the corset structure garment stability. A layered tulle skirt was then constructe

2.4 System Integration

The fnal assembly integrated biometric sensing, wireless co construction into a single wearable prototype. Heart rate transmitted wirelessly through the ESP32-C3 modules, proce servo-driven foral movement.

3 Results

3.1 Hardware Performance

Servo durability testing included running the fower mechani separate tests were run and demonstrated mechanical failure tuation cycles. Under continuous operation, this correspon two hours before actuator degradation and burn-out was obse continuous readings before degradation occurred. The micro continuous testing.

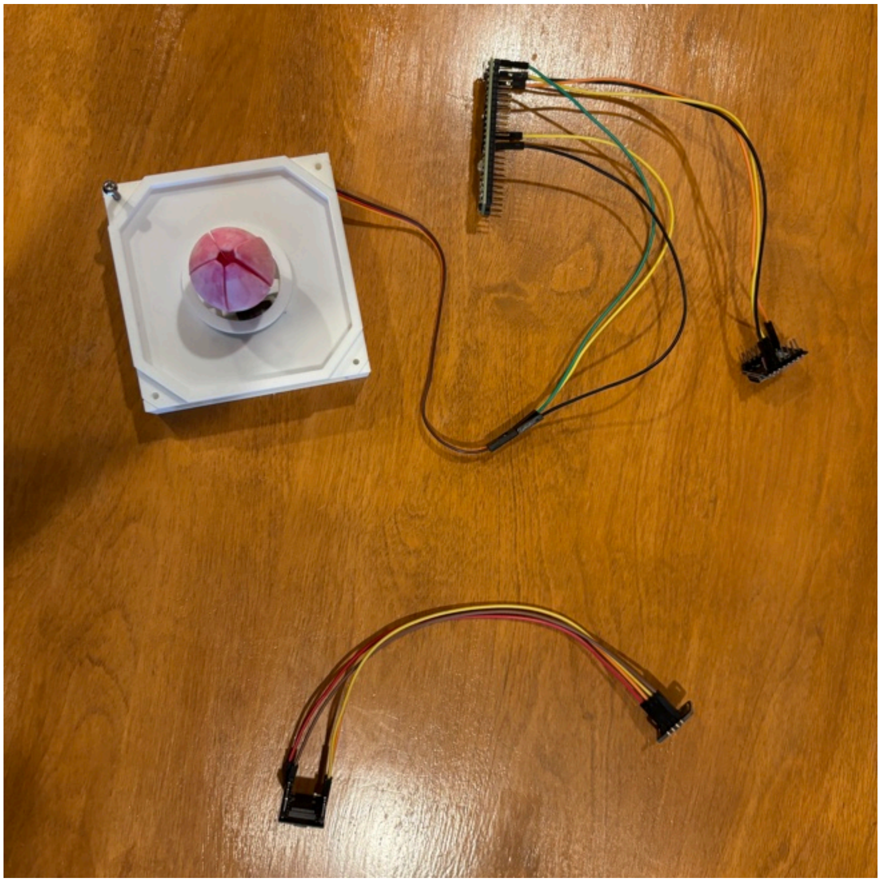


Figure 9: Set-up of integration. Servo is housed inside of th

3.2 Wireless Communication Performance

Wireless communication between the transmitting and receivi using the ESP-NOW protocol. Sensor data packets were transmi observable communication loss under short-range conditions

3.3 Final Wearable Prototype

The completed garment successfully integrated biometric se ware, structural support components, and soft textile cons heart rate input produced corresponding kinetic movement in ful physiological-to-mechanical translation.

4 Conclusions

This research demonstrated the feasibility of integrating b tion, robotic actuation, and couture garment construction i translating physiological data into dynamic physical expre

The system successfully detected cardiovascular activity, t corresponding servo-actuated floral movement within a fully

Several limitations were identifed during development. The tem were strongly infuenced by the quality and reliability o

particularly the servo motors and heart monitor under repea
availability and fnancial constraints limited opportuniti
tion materials, and repeated hardware iteration.

Future work may focus on improved actuator durability, ligh
inputs, and increased long-term wearability for practical d

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